

Wuhan University 2025–2026 Spring Introduction to Data Science Final Exam

(Total: 100 points Exam duration: 120 minutes)

1. (20 points)

Define the target function as

$$f^* \in \arg \min_{\text{measurable } f} \mathcal{R}(f) = \mathbb{E}_{(X,Y) \sim P} [\ell(f(X), Y)].$$

Show that:

1. $f^*(x) = \arg \min_{a \in \mathcal{Y}} \mathbb{E}[\ell(a, Y) \mid X = x]$.
2. For regression: $\mathcal{Y} = \mathbb{R}$, $\ell(a, y) = \frac{1}{2}(a - y)^2$, prove that $f^*(x) = \mathbb{E}[Y \mid X = x]$.
3. For classification: $\mathcal{Y} = [K] := \{1, 2, \dots, K\}$, $\ell(a, y) = \mathbf{1}_{a \neq y}$, prove that $f^*(x) = \arg \max_{k \in [K]} \mathbb{P}(Y = k \mid X = x)$.

2. (20 points)

Let $f(x) = x^2$ on $[0, 1]$, construct a neural network with ReLU activation to approximate it. And show the dependence of the depth and width on the approximation accuracy.

3. (20 points)

Prove the **Dudley's entropy formula**: Assume $0 \in \mathcal{U}$ and the diameter of $\mathcal{U}$ is less than $\mathcal{B}$, i.e., $\|u\|_{L^\infty(\Omega)} \leq \mathcal{B}, \forall u \in \mathcal{U}$. Then

$$\|R_n\|_{\mathcal{U}} \leq \inf_{0 < \delta < \mathcal{B}} \left(4\delta + \frac{12}{\sqrt{n}} \int_{\delta}^{\mathcal{B}} \sqrt{\log(2\mathcal{N}_n(\epsilon, \mathcal{U}, \ell_\infty))} d\epsilon \right).$$

4. (20 points)

Consider the linear model $Y = X\theta^* + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ and $n \ll d$. Define the Lasso estimator:

$$\hat{\theta}_L = \arg \min_{\theta \in \mathbb{R}^d} \left\{ \frac{1}{2n} \|Y - X\theta\|_2^2 + \lambda \|\theta\|_1 \right\}.$$

Let

$$\gamma = \inf_{v \in C} \frac{v^\top \frac{X^\top X}{n} v}{\|v\|_2^2} > 0,$$

with $C = \{v : \|v_{A^*}\|_1 \leq 3\|v_{A^*}\|_1\}$ and $A^* = \text{supp}(\theta^*)$. Set

$$\lambda = 2\sigma \sqrt{\frac{2 \ln(2d/\delta)}{n}}.$$

Prove that, with probability at least $1 - \delta$,

$$\|\hat{\theta}_L - \theta^*\|_2 \leq \frac{6}{\gamma} \sqrt{2 \ln(2d/\delta)} \sigma \sqrt{\frac{|A^*|}{n}}.$$

5. (20 points)

Assume that

$$\langle \nabla f(x), x - x^* \rangle \geq \gamma(f(x) - f(x^*)), \quad \gamma > 0.$$

Prove that $f(x^t) - f(x^*) \leq \mathcal{O}(1/t)$, where x^t denotes the t -th iterate of gradient descent.